

2ND INTERNAL ASSESSMENT - 2026

Topic: Polynomials Class- IX Session: 2026-27

Sub.- Mathematics

Time- 60 Minutes

F.M.- 30

Section A: Multiple Choice Questions (MCQ)

1. Which of the following is not a polynomial? [1]
(a) $x^2 + 5x + 6$ (c) $\sqrt{x} + 2$
(b) $3x + 7$ (d) 9
2. Read the two statements and choose the correct option. [1]
(I) The constant 8 is a polynomial of degree 0.
(II) Every constant polynomial is also a linear polynomial.
(a) Only I is true (c) Both I and II are true
(b) Only II is true (d) Neither I nor II is true
3. A chess club charges ₹200 as joining fee plus ₹50 per match. The expression for the total amount paid after m matches is: [1]
(a) $50m + 200$ (c) $250m$
(b) $200m + 50$ (d) $200 - 50m$
4. The value of the quadratic polynomial $7s^2 - 4s + 6$ at $s = -3$ is: [1]
(a) $63 + 12 + 6 = 81$ (c) $-63 + 12 + 6 = -45$
(b) $63 - 12 + 6 = 57$ (d) $63 - 4 + 6 = 65$
5. The line $y = -3x + 4$ cuts the y-axis at the point: [1]
(a) $(0, -3)$ (c) $(4, 0)$
(b) $(0, 4)$ (d) $(0, -4)$
6. The line $y = ax + b$ passes through the origin. Then: [1]
(a) $a = 0$ (c) $a = b$
(b) $b = 0$ (d) $a + b = 0$
7. In a linear pattern, the difference between any two consecutive terms is [1]
8. If a quantity decreases at a constant rate, it follows Linear [1]
9. Which of the following statements is NOT true? [1]
(a) Degree of a zero polynomial is not defined.
(b) The degree of a constant polynomial is always 0.
(c) A polynomial of degree 2 is called binomial.
(d) A polynomial of degree 3 is called cubic polynomial.
10. A rally starts with 120 members. Each hour, 9 members drop out. The number of members at the end of the n th hour is: [1]

(a) $120n - 9$

(b) $9n + 120$

(c) $120 - 9n$

(d) $129 - n$

Section B: Answer the following questions

1. Identify whether each of the following is a polynomial. Justify in one line: [2]

(a) $3\sqrt{x} + 4$

(b) $\frac{2}{x} + x$

2. Find a linear polynomial whose graph is parallel to $y = 5x - 7$ and passes through $(0, 4)$. [2]

3. The graph of a linear polynomial $p(x)$ passes through the points $(1, 5)$ and $(3, 11)$. [6]

(a) Find the polynomial $p(x)$.

(b) Find the coordinates where the graph of $p(x)$ cuts the axes.

(c) Draw the graph of $p(x)$ and verify your answers.

4. A telecom company charges a fixed monthly fee plus an additional cost per GB of data used. Priya observes: when she used 10 GB her bill was ₹350; when she used 20 GB her bill was ₹550. Let $y = \text{bill}$, $x = \text{data used (GB)}$, with $y = ax + b$. [6]

Based on the given information, answer the following questions:

(a) Find a and b .

(b) Interpret the values of a and b in this real-life context.

(c) If Priya wants to keep her bill below ₹400, what is the maximum data she can use?

5. Aarti writes a polynomial $p(t) = 4t^3 - t^2 + 6$ to model the temperature change of a chemical reaction at time t hours. [4]

(a) State its degree and type.

(b) Find $p(0)$ and $p(2)$.

(c) Identify the constant term.